

Instructor Key: Day 7 Mastery Quiz: Ordered Validation

Ordered Validation
Key for the Week 2 student quiz package. Keep instructor-facing.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 7 mastery quiz; quiz, not workbooklet.
Content anchor	Built from the locally verified Week 2 workbooklet content anchor for sensor-kit checkout and release routing.
Difficulty posture	Shorter and sharper than the workbooklet. It asks for trace, route, debug, test, and transfer evidence.
Student-surface rule	Student PDF hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.

2 Reference code

```
1 damage_note = ""
2 charge_percent = 83
3 calibration_label = "expired"
4 borrower_initials = "MK"
5
6 if damage_note != "":
7     route = "repair bench"
8 elif charge_percent < 80:
9     route = "charge station"
10 elif calibration_label != "current":
11     route = "calibration bench"
12 elif borrower_initials == "":
13     route = "borrower log needed"
14 else:
15     route = "release kit"
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Support move
1	1. first condition	damage_note != "" is checked first; it is false because damage_note is blank.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	2. second condition	charge_percent < 80 is checked second; 83 < 80 is false.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	3. third condition	calibration_label != "current" is true because the label is expired.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	4. route	"calibration bench".	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	5. skipped later rule	borrower_initials == "" is skipped after the calibration rule is true.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	6. damage priority	"repair bench" because damage is the first true route and stops the chain.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	7. charge priority	"charge station" because low charge appears before calibration.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	8. else route	"release kit".	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	9. concept	In if/elif/else, the chain stops at first true; independent checks could record multiple issues.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	10. evidence sentence	Secure answer names false damage, false charge, true expired calibration, route calibration bench, and skipped later checks.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.

4 Mastery interpretation

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10	Student names the route evidence, controlling condition, and transfer/debug implication.	Move to the next daily packet or Exam 1/Week 3 bridge as appropriate.
Developing	5–7 / 10	Student can get final route but may not preserve condition order or expected/actual evidence.	Use a route ledger: condition, value, branch, action, reason.
Needs support	0–4 / 10	Student guesses route from scenario language or final output without tracing.	Return to first condition, true/false result, branch selected, and support note.

5 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores.